

★ WEEK 5 · PART 3 of 3

Volatility in Practice

Value-at-Risk, Back-Testing, and Pairs-Trading Bands

GARCH → VaR & Expected Shortfall · violation back-tests · the spread's breathing bands

A volatility forecast is only useful once it drives a decision. Part 3 turns GARCH into a risk number, tests that number honestly, and launches the pairs-trading thread that runs through the course.

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PART 3 · IN PRACTICE

Turning Volatility Into Decisions

From a GARCH forecast to Value-at-Risk and Expected Shortfall, back-testing those numbers, and using conditional volatility to build a pairs-trading strategy.

From a Forecast to a Decision

Part 2 built the model; Part 3 puts it to work

1

We can forecast conditional volatility

A fitted GARCH gives today's risk and its forecast path — the size of tomorrow's moves, even though their direction is unknowable.

2

Now we convert risk into numbers and rules

That single volatility feeds Value-at-Risk, Expected Shortfall, position sizing, and trading bands — the everyday tools of a risk desk.

3

And we start the pairs-trading thread

Conditional volatility on a spread gives 'breathing' bands — Stage I of the pairs strategy that recurs through the rest of the course.

Part 3 in one line: a good volatility forecast powers VaR, back-tests, and trading bands.

The GARCH Workflow

From returns to a validated volatility model

1 • Mean

keep simple

Fit a minimal mean model; work with the residuals.

2 • Test

ARCH-LM

Confirm the residuals show volatility clustering.

3 • Fit

GARCH family

Estimate GARCH/GJR/EG ARCH, normal or Student-t.

4 • Validate

residuals

Standardized residuals: no ARCH, right tails.

5 • Forecast

and use

Forecast volatility → VaR, ES, and bands.

Steps 1–4 are Part 2; step 5 is where this part begins — turning the forecast into decisions.

From Volatility to a Risk Number

Making the forecast actionable

A conditional volatility is abstract; a dollar loss limit is not. Value-at-Risk converts today's forecast volatility into a concrete number a risk manager can act on.

What managers need

- A single figure: how bad could a normal-day loss get?
- One that reflects TODAY's risk, not a long-run average.
- One that regulators and desks can monitor daily.

How GARCH supplies it

- It forecasts the size of tomorrow's move directly.
- Scale that by a quantile and you have a loss limit.
- The limit tightens in calm and widens in storms.

Value-at-Risk (VaR)

The loss you should not exceed on a normal day

VaR at 99% is the loss level that should be breached on only 1 day in 100. It is a quantile of the loss distribution — and its width is driven by volatility.

How to read it

- '99% 1-day VaR = 2%' means: worse than -2% on ~1 day in 100.
- A confidence level (95%, 99%) and a horizon (1 day) define it.
- It is a threshold, not the worst case.

Its famous flaw

- VaR says nothing about HOW bad the breach days are.
- Two portfolios with equal VaR can have very different tails.
- Expected Shortfall fixes exactly this (a few slides on).

Value-at-Risk From a GARCH Forecast

Volatility times a quantile

$$\text{VaR}_t(\alpha) = -(\mu + \sigma_t q_\alpha) \quad q_\alpha = \alpha\text{-quantile of } z_t$$

The pieces

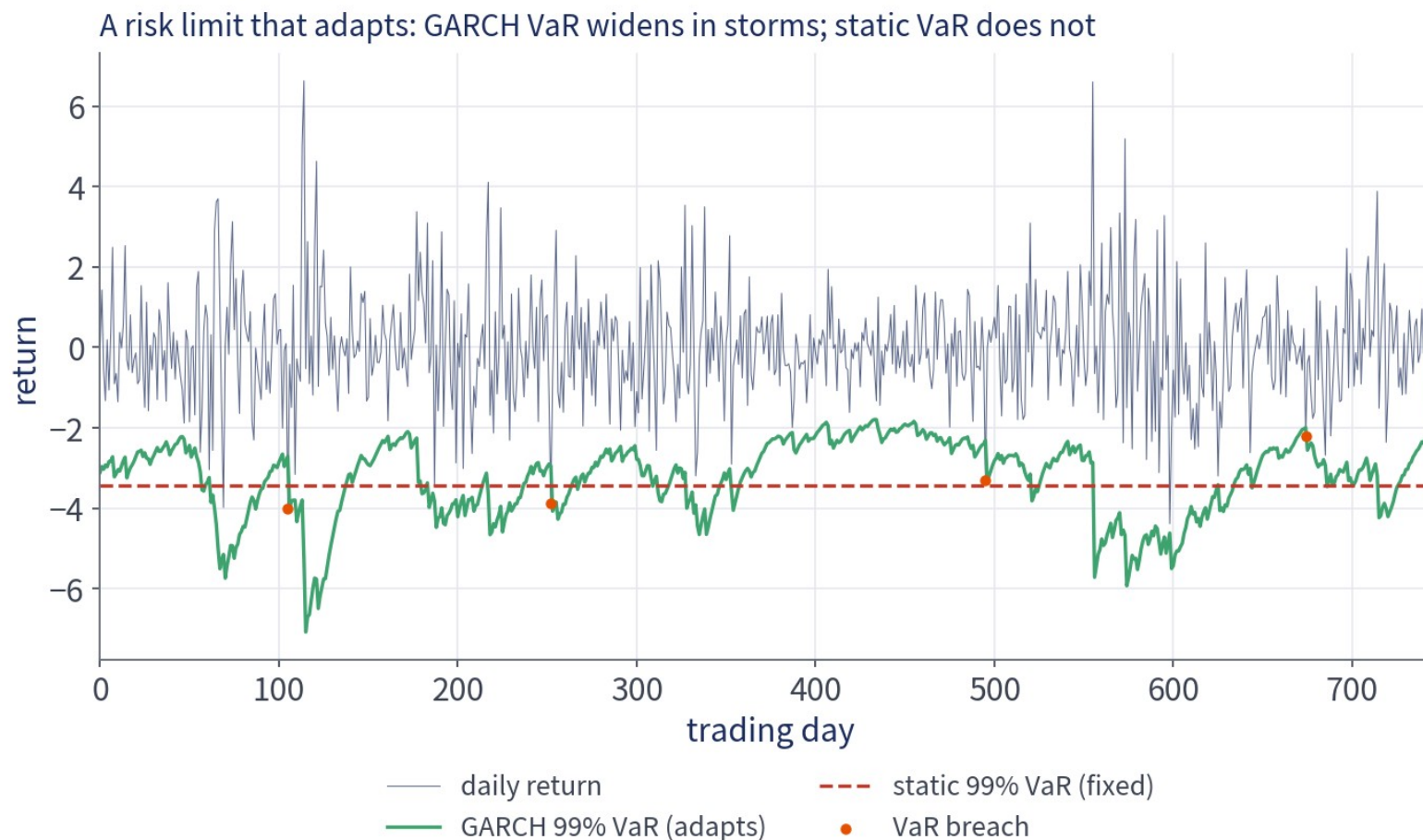
- μ is the (small) expected return; σ_t is today's GARCH volatility.
- q_α is the α -quantile of the standardized innovations.
- Use the Student-t quantile if you fitted fat tails.

Why it adapts

- σ_t changes daily, so the VaR breathes with the market.
- After a shock, σ_t rises and the limit widens automatically.
- This is the whole advantage over a static VaR.

A Risk Limit That Adapts

The GARCH VaR widens in turbulent stretches and tightens in calm ones; a static VaR stays flat and is breached in clusters exactly when risk is highest.



Why a Static VaR Fails

A single number for all seasons is dangerous

The static trap

- A fixed VaR is too loose in calm — it overstates day-to-day risk.
- And far too tight in a crisis — it understates the real danger.
- Its breaches cluster in storms, exactly when it matters most.

The GARCH remedy

- The limit tracks the current regime, day by day.
- Breaches spread out evenly instead of clustering.
- That even spacing is the signature of a well-calibrated VaR.

Expected Shortfall

The average loss WHEN things go wrong

$$ES_t(\alpha) = \mathbb{E} \left[L_t \mid L_t > \text{VaR}_t(\alpha) \right] \quad (\text{expected shortfall})$$

What it adds

- ES is the average loss GIVEN that VaR is breached.
- It looks INTO the tail, not just at its edge.
- It is now the regulatory standard (Basel) for market risk.

Why it is better

- It distinguishes portfolios that VaR rates identically.
- It is coherent — it rewards diversification properly.
- Compute it from the same GARCH forecast and tail quantiles.

Back-Testing VaR

Did reality respect your risk number?

A 99% VaR should be breached about 1% of the time — no more, no less. Back-testing counts the actual breaches and asks whether that rate is statistically right.

How it works

- Compare each day's actual loss to that day's VaR.
- Count the breaches over a long window.
- Compare the count to the expected number, $\alpha \cdot N$.

What to look for

- Too many breaches → the VaR is too optimistic.
- Breaches that CLUSTER → the model misses changing risk.
- Ideally: the right count, evenly spaced.

The Kupiec Test

Is the breach rate statistically correct?

count violations: expect αN ; Kupiec LR test: is the breach rate $= \alpha$?

What it checks

- Under a correct model, breaches occur at rate α .
- Kupiec is a likelihood-ratio test on the observed count.
- A small p-value rejects the VaR model as miscalibrated.

Beyond the count

- Kupiec tests the NUMBER of breaches.
- Independence tests check they do not CLUSTER.
- A good VaR passes both — right count, no clustering.

Back-Testing, Visualized

Cumulative breaches should track the expected 1% line. The GARCH VaR stays close to it; the static VaR drifts above — too many breaches, clustered in storms.



What a Good VaR Model Looks Like

Two things must both be true

The two criteria

- Correct frequency: about α breaches, no systematic bias.
- Independence: breaches do not bunch together in time.
- A model can pass one and fail the other — check both.

The GARCH advantage

- Because it adapts, its breaches are spread through time.
- It reacts to a new regime within a day or two.
- That responsiveness is why it back-tests so well.

Pairs Trading Starts Here

The strategy that runs through the whole course

Recall cointegration (Week 2): two assets whose spread mean-reverts. Trading that reversion needs a threshold — and conditional volatility is what makes the threshold intelligent.

The setup

- Find a cointegrated pair; form its stationary spread.
- Enter when the spread is 'stretched', exit as it reverts.
- But how far is 'stretched'? That depends on the spread's risk.

Where GARCH enters

- Model the spread's conditional volatility with GARCH.
- Set the entry bands at a multiple of that volatility.
- The bands widen in volatile regimes, tighten in calm ones.

The Spread and Its Volatility

Volatility is not constant here either

The spread's dynamics

- The spread mean-reverts (cointegration) but its risk varies.
- Some periods it is quiet; others it swings violently.
- A fixed band ignores this — and fires at the wrong times.

Modeling it

- Apply GARCH to the spread's changes, just as for returns.
- Get a conditional volatility for the spread each day.
- This is the input to the trading bands.

GARCH Conditional Bands on the Spread

Thresholds that scale with the spread's risk

$$S_t = P_t^A - \beta P_t^B, \quad \sigma_{S,t}^2 \text{ modeled by GARCH,} \quad \text{bands} = \pm k \sigma_{S,t}$$

Building the bands

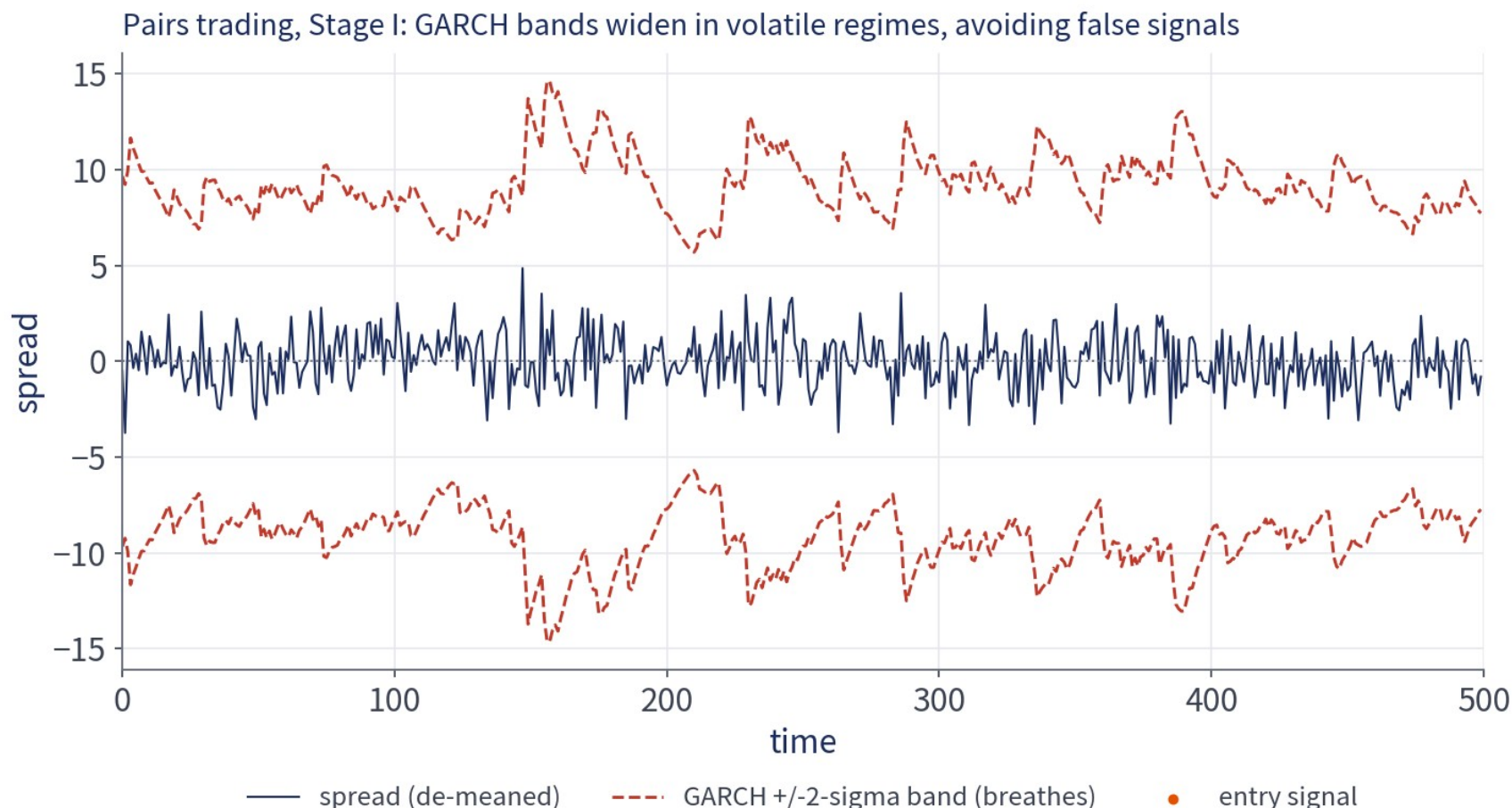
- Form the spread from the cointegrating relationship.
- Model its conditional volatility $\sigma_{S,t}$ with GARCH.
- Set entry/exit bands at $\pm k$ times that volatility.

Why it is smarter

- In calm, tight bands catch small, reliable reversions.
- In turbulence, wide bands avoid premature entries.
- The threshold adapts to the regime automatically.

Pairs Trading, Stage I

The spread with GARCH $\pm 2\sigma$ bands that breathe: they widen in volatile regimes and tighten in calm, so entry signals fire when a move is genuinely unusual.



Why Breathing Bands Beat Fixed Ones

Fewer false signals, better timing

Fixed bands

- Too tight in volatile regimes → constant false entries.
- Too wide in calm regimes → missed reversions.
- They assume a risk level that simply is not constant.

GARCH bands

- Scale the threshold to the spread's current volatility.
- Signals fire only when a move is unusual for TODAY.
- Cleaner entries, and fewer whipsaws in stormy periods.

The Pairs-Trading Roadmap

Where this thread goes from here

Stage I gives us intelligent bands. The later stages — signal generation, position sizing, and portfolio construction — build on this same volatility-aware foundation.

Done here (Stage I)

- Cointegration finds the pair (Week 2).
- GARCH gives the spread's conditional volatility.
- Breathing bands define when the spread is stretched.

Coming later

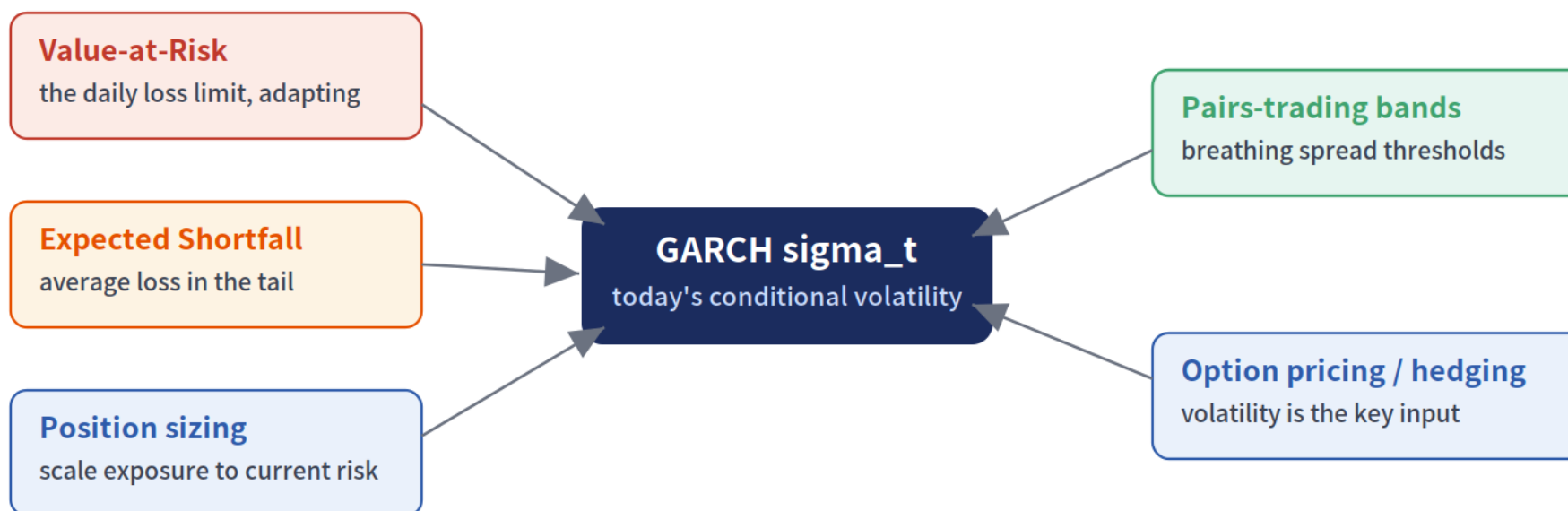
- Turning band crossings into disciplined entry/exit rules.
- Sizing positions by risk, using the same volatility.
- Combining many pairs into a diversified book.

What a Volatility Forecast Unlocks

One GARCH conditional volatility feeds Value-at-Risk, Expected Shortfall, position sizing, option hedging, and pairs-trading bands — the whole risk toolkit.

What a Volatility Forecast Unlocks

One GARCH conditional variance feeds every downstream risk and trading decision.



The theme of the week: you cannot forecast the return, but a good volatility forecast powers nearly every risk decision.

One Asset at a Time Isn't Enough

Why we need to go multivariate

GARCH models each series' volatility alone. But portfolios and pairs depend on how assets move TOGETHER — and correlations, like volatilities, change over time.

The limitation

- Univariate GARCH ignores cross-asset relationships.
- Portfolio risk depends on time-varying correlations.
- Even our pair needs the joint dynamics of two series.

Where the course turns next

- Multivariate models let volatilities and correlations co-move.
- Vector autoregression (VAR) models several series jointly.
- That is exactly where Week 6 begins.

VaR Back-Test and Pairs Bands

Direct Claude Code from risk numbers to a trading signal

GOAL Build a GARCH VaR, back-test it, then form GARCH bands on a cointegrated spread and mark entry signals.

[INPUT] Prompt · Claude Code

Write and run `vol_practice.py` that:

1. fits a $\text{GARCH}(1,1)$ -t to a stock's returns and computes a rolling 99% 1-day VaR;
2. back-tests it: count breaches, run the Kupiec test, and compare to a static VaR;
3. for a cointegrated pair, models the spread's volatility with GARCH and plots ± 2 -sigma bands with entry signals.

Report the breach rate, the Kupiec p-value, and how many signals the breathing bands give versus fixed bands.

How to work with it

- A good VaR: breach rate $\approx 1\%$, Kupiec not rejected.
- Static VaR usually breaches too often, in clusters.
- Breathing bands should give cleaner, fewer signals.
- This closes Week 5 and sets up Week 6.

Deliverable: `vol_practice.py` + a VaR back-test report and a pairs-band chart.

Three Questions Before You Finish

If you can answer these, Week 5 has landed

Q1

Your 99% VaR is breached on 4% of days, in clusters. What's wrong?

VaR too optimistic / static

Q2

Why does Expected Shortfall improve on VaR?

Looks into the tail

Q3

Why do GARCH bands beat fixed bands for a pairs spread?

Adapt to spread volatility

All three follow from time-varying risk, tail measures, and adaptive thresholds.

Three Pitfalls to Avoid

Where volatility-in-practice goes wrong

Common mistakes

- "A single VaR number captures all my risk."
- "If VaR looks fine, no need to back-test."
- "Fixed bands are simpler, so use them for pairs."

The reality

- VaR ignores tail severity — pair it with Expected Shortfall.
- Always back-test: check breach count AND clustering.
- Fixed bands misfire; GARCH bands adapt to the regime.

Week 5 in Three Ideas

Pulling the whole week together

1

Model the variance, not the mean

Part 1: returns are unpredictable in direction but their variance is not. ARCH ties conditional variance to recent squared shocks — clustering, formalized.

2

GARCH is the workhorse

Part 2: a memory term (β) captures long volatility memory in three parameters; Student-t and GJR/EGARCH add fat tails and the leverage effect.

3

Volatility drives every risk decision

Part 3: a GARCH forecast becomes VaR, Expected Shortfall, and breathing pairs-trading bands — validated by back-tests.

Next — Week 6: Multivariate time series and VAR, modeling how assets move together.